MUCHUKUNTLA BRAHMANAIDU

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EDUCATION

Annamacharya Institute of Technology and Sciences, Kadapa ***Dec. 2021– Apr 2025***
• Bachelor of Engineering in Computer Science *CGPA: 7.9 / 10*
• **Coursework:** Expertise in Statistical Modeling, Machine Learning/Deep Learning, and Big Data *Andhra Pradesh*

RVSSV Junior College, Khajipeta ***Jun. 2019– Apr 2021***
•Intermediate (MPC) *12th GPA: 8.3*
• **Coursework**: Mathematics, Physics, Chemistry *Andhra Pradesh*

Experience

Machine Learning Engineer Intern | *Zithara.AI (Propel5000)* | ***Jan 2025 – Apr 2025***

- Visual Search Engine: Built a 'Search-by-Image' feature for a MERN-stack e-commerce app, enabling users to find similar jewellery from a 5,000+ SKU catalog.
- Computer Vision: Developed a PyTorch (ResNet50) pipeline to generate vector embeddings, capturing complex gold textures for accurate retrieval.
- Performance & Integration: Implemented FAISS for similarity search (<200ms latency) and integrated the Python microservice with Node.js/React for real-time inference.

Projects

Automated MLOps Pipeline for Model Lifecycle Management | Python, MLflow, CI/CD, Docker, DVC | ***Apr 2025 – May 2025***

- Built a scalable ML microservice covering the full lifecycle: data ingestion, feature engineering, training, and real-time inference via REST APIs.
- Containerized the application using Docker to ensure environment consistency and deployed via FastAPI for high-performance model serving.
- Implemented production-grade logging and monitoring to track system health and inference latency, ensuring API reliability under load

LLM-Powered Intelligent Assistant with RAG | Python, LangChain, OpenAI, Pinecone, FastAPI | ***Nov 2024 – Jan 2025***

- Developed a Generative AI application using Retrieval-Augmented Generation (RAG) to query custom knowledge bases, reducing manual information retrieval time by 40%.
- Designed robust prompt engineering templates and guardrails to minimize hallucinations, ensuring 95%+ factual consistency in generated responses.
- Optimized semantic search accuracy by implementing hybrid retrieval strategies and fine-tuning embedding chunk sizes for context-aware answers.

Transformer-Based Semantic Search | Python, BERT, Hugging Face, PyTorch | ***Aug 2024 – Oct 2024***

- Fine-tuned pre-trained BERT models for complex NLP tasks including multi-label classification and semantic search, achieving an F1-score of 0.89.
- Engineered an inference pipeline using Hugging Face transformers, outperforming traditional TF-IDF approaches by capturing deep contextual nuances.
- Optimized model performance and reduced inference latency by 30% through dynamic quantization and efficient tokenization strategies

TECHNICAL SKILLS

• **Languages & Core:** Python, SQL (PostgreSQL), Pandas, NumPy
• **Machine Learning & DL:** PyTorch, Scikit-learn, TensorFlow, XGBoost, FAISS
• **MLOps & Engineering:** FastAPI, Docker, MLflow, Git, AWS
• **Generative AI & NLP:** LangChain, Hugging Face, RAG, BERT, LLMs
• **Data Viz & Analytics:** Power BI, Matplotlib, Seaborn, Prophet (Time-Series)

CERTIFICATIONS

• Google Data Analytics Professional Certificate • Brainovision Data Science Internship Certificate
• Udemy Certificates(SQL | POWER BI | ML | DL | DATA SCIENCE | LLM)